Subject:

Summarize :-

(i) Define a state space that contain all possible cond. of the relevant objects.

(ii) Specify one or more state that describe possible situation from which the problem solving may start (i.e. initial state)

(iii) Specify one or more state that will be accepted as goal state.

(iv) Specify a set of rules that describe the actions available.

Production System :-

→ Set of rule, left side specify the applicability of the rule and right side specify operation to be performed.

→ One or more knowledge database

→ Control strategy :- specify the order in which rules will be compared in the database.

→ A way of resolving conflicts when several rules match at once.

→ Rule applier

Subject: * Control strategies

* what next rule i should apply so that my goal state is achieved.

→ which rule to apply next for achieving the goals.

(i) It should cause motion

(ii) Control strategy must be systematic

systematic
- → global motion
- → local motion
- → order
  - → BFS
  - → DFS

BFS

↳

- create a variable NODE_LIST and set it to initial state

- Until a Goal state is found or NODE_LIST is empty do

  (a) Remove the first element from NODE_LIST and call it 'E'.
  NODE_LIST was empty then QUIT

  (b.) For each way the rule can match the state described in 'E'

Subject:

(i) apply the rule / /
to generate a new state
(ii) If new state is goal state
then QUIT
(iii) If new state is not goal
state add this state to
NODE_LIST

BFS advantages

(i) BFS will not get trapped exploring
a blind alley

(ii) If there is a solution, BFS gurantees you
to give the sol^n.

Furthermore if there are
multiple sol^n, you will get the minimal
sol^n.

DFS :-

1. If initial state is GOAL state then QUIT.

2. Do the following UNTIL SUCCESS or FAILURE is
designated :-
(a) Generate a successor 'E' of the intial
State. If no successor then QUIT.

Subject: / /

(b) Call depth first search with E as initial state.

(c) If success is returned signal success otherwise continue in LOOP.

Advantages:-

(i) Less m/m.

(ii) DFS may find the solⁿ without exploring the much of the state space.

To Do

→ Travelling salesman problem apply this topic